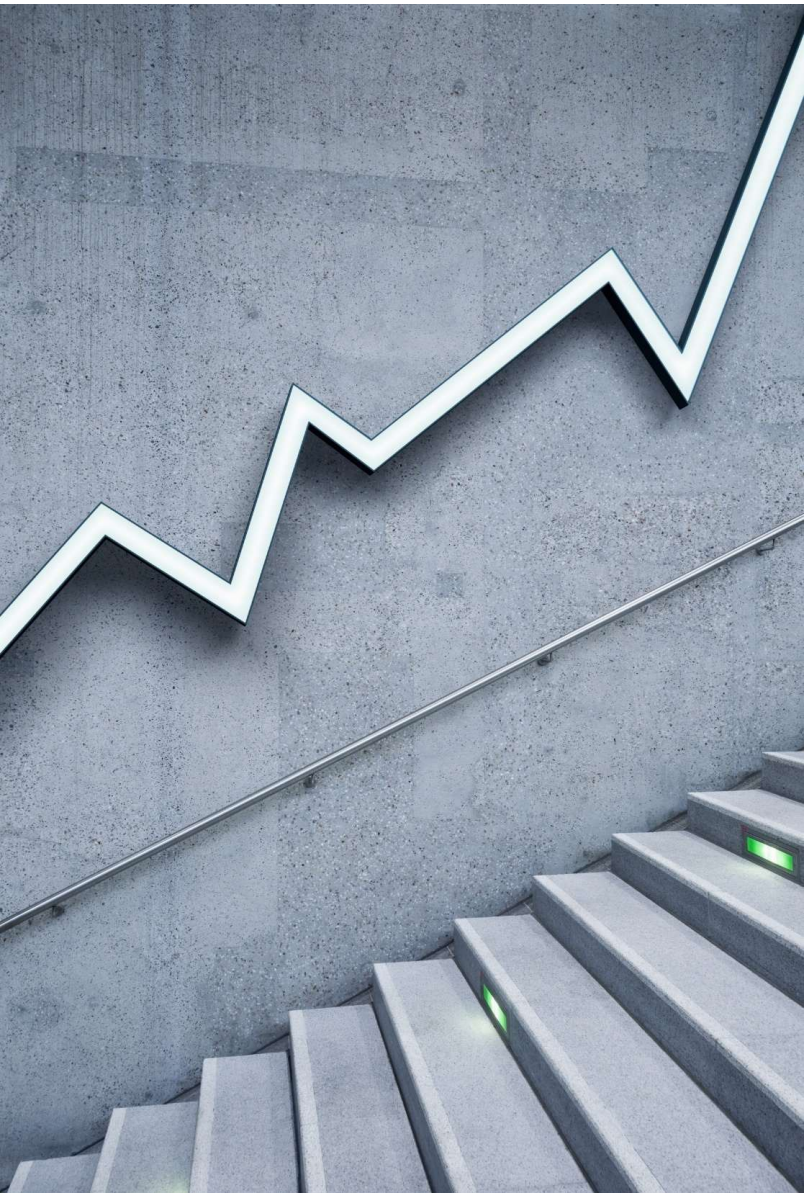


Learning Python

PRAVEEN NAIR

Part 1

PYTHON FUNDAMENTALS

Introduction to Python

1. Invented in the Netherlands, early 90s by Guido van Rossum named after Monty Python Flying Circus series
2. Open sourced from the beginning
3. Scalable, object oriented and functional from the beginning
4. Increasingly popular
5. Strong Libraries

What is Python used for?

1. Beginner-friendly programming language
2. Data Analysis and Machine Learning
3. Web & Desktop Application Development
4. Automation or Scripting
5. Programming Internet of Things (IoT) devices

Implementations of Python

CPython – the standard and most widely used implementation.

PyPy – a faster implementation with a Just-In-Time (JIT) compiler.

Jython – runs on the Java Virtual Machine (JVM).

IronPython – runs on the .NET framework.

MicroPython – designed for microcontrollers and embedded devices.

Installing Python…

1. Visit https://www.python.org/downloads/
2. Click on Download Python button
3. Run the installer (make sure to tick add path checkbox)

Installing Visual Studio Code…

1. Visit https://code.visualstudio.com/download
2. Click on Download button
3. Run the installer



Variables

We use variables to temporarily store data in computer's memory.

price = 10 (Integer) #type(price)

rating = 4.9 (Python represent float values as 64-bit "double-precision" values so no separate type as double)

course_name = 'Python for Beginners' (String)

is_published = True (Boolean)

Data Structures – List, Tuple, Dictionary, Set, Array (A data structure is a specialized format for organizing, processing, retrieving and storing data.)

……

Type function, multiple assignment, variable naming rule

Multi-line String

To define a multi-line string, we surround our string with tripe quotes.

```
msg = '''
Dear Friend,
How are you? Hope all is well with you.
Thanks,
MyName
'''
print(msg)
```

Type Conversion

```
birth_year = input("Enter your birth year")
age = 2022 - int(birth_year)
print(age)
---------------
 float()
 Eval()
```

Printing Variables

```
name="John"

print(name)
```

Receiving Input

```
name=input("Enter your name")

print(name)
```

Concatenate Strings

```
first_name = "John "

last_name = "Smith "

print(first_name + last_name)
```

Formatted String Literals (f-string)

```
name=input("Enter Name ")
favorite_color=input("Enter Favorite Colour ")
msg = f'{name} likes {favorite_color}’
print(msg)
```

Comments

```
birth_year = input("Enter your birth year")

age = 2022 - int(birth_year)

# print(age)

Print(birth_year)

---------

How to comment multiple lines?
```

Arithmetic Operations

```
+
-
*
/  returns a float
//  returns an int
%  returns the remainder of division
** exponentiation
```

Comparison Operators in Python

```
Equals: a == b

Not Equals: a != b

Less than: a < b

Less than or equal to: a <= b

Greater than: a > b

Greater than or equal to: a >= b
```

Logical Operators in Python

```
and: (a < b) and (a < c)

or: (a < b) or (a < c)

Not: not(a > c)
```



Python Escape Characters

```
txt="This is John\'s book." # \' means '

print(txt)

txt="This is a \n dummy text." # new line

print(txt)

txt="Name \t City \t Age" # tabspace

print(txt)
```

if...statement

```
a=10
b=15
if(a>b):
    print("a is big")
else:
    print("b is big")
```

if…elif…statement

```
a=10
b=10
if(a>b):
    print("a is greater")
elif(b>a):
    print("b is greater")
else:
    print("Both a and b are equal")
```

Switch/Match statement…

```
subject="HTML"
match subject:
    case "HTML":
        print("HyperText Markup Language")

    case "CSS":
        print("Cascading Style Sheet")
    case _:
        print("Unknown")
```

Printing with end parameter

By default, the print function ends with a newline.

Passing the whitespace to the end parameter (end=‘ ‘) indicates that the end character has to be identified by whitespace and not a newline.

```
----------------------

print("Hello", end=' ')

print("World")
```

While loops

```
i = 1
while i < 5:
    print(i)
    i += 1
------------------
Print Series –
5,10,15,20,25,30.....
```

While loops – break statement

```
secret_number=5
guess_count=0
guess_limit=3

while(guess_count < guess_limit):
    guess = int(input("Guess the number: "))
    guess_count = guess_count + 1
    if (guess==secret_number):
        print("You Win!")
        break
else:
    print("Better luck next time!")
```

While loops – continue statement

```
i=0
while (i<5):
  i = i + 1
  if (i==3):
    continue
  print(i)
```

While loops – pass statement

```
i=0
while (i<5):
  i = i + 1
  if (i==3):
    pass
  else:
    print(i)
```

For loops

```
for i in range(5): # iterates over a sequence
    print("Hello")
--------------
```

- `range(5)`: generates 0, 1, 2, 3, 4
- `range(1, 5)`: generates 1, 2, 3, 4
- `range(1, 5, 2)`: generates 1, 3

For loop with else

```
n = 23
for i in range(2,n):
    if n % i == 0:
        print("Not a prime number")
        break
else:
    print("Prime Number")
.....................
```

Try with while..else also

Nested For loop

```
for i in range(5):
  for j in range(5):
      print(j)
```

Assignment: Student Information and Marks

Write a Python program that accepts the following details from the user using **Scanner**:

1. Student name
2. Student age
3. Student roll number
4. Marks in Java
5. Marks in Python
6. Marks in Database

Calculate and display:

1. Student details
2. Total marks
3. Average marks

No if or loops or functions

Assignment: Employee Salary Calculator

Write a Python program that accepts the following details from the user:

1. Employee name
2. Employee ID
3. Basic salary
4. HRA
5. DA

Calculate and display:

1. Employee details
2. Gross salary = Basic Salary + HRA + DA
3. Annual salary = Gross Salary × 12

No if or loops or functions

Assignment: Shopping Bill Calculator

Write a Python program that accepts:

1. Customer name
2. Product name
3. Product price
4. Quantity

Calculate and display:

1. Total price
2. Discount of 10%
3. Final bill amount after discount

No if or loops or functions

Student Result & Grade Calculator

Create a Python program that accepts a student's marks in **5 subjects**, calculates the total and percentage, and determines:

Pass or Fail

Grade

Division

Scholarship eligibility

Note: Do not use loops, arrays, methods, classes beyond the main class, or switch

Use if statement

Requirements

The student enters:

Student name

Java marks

Python marks

Database marks

Web Technology marks

Aptitude marks

Each subject is out of 100.

Condition 1 - Validate marks

If any mark is:

Less than 0 ⟶ Invalid marks

Greater than 100 ⟶ Invalid marks



Condition 2 - Pass/Fail

A student passes only if **each subject is at least 40.**

Otherwise:

Result: FAIL

Reason: One or more subjects below 40

Condition 3 - Grade

Percentage	Grade
90–100	A+
80–89	A
70–79	B
60–69	C
50–59	D
40–49	E

Condition 4 - Division

Percentage	Division
75+	Distinction
60–74	First Class
50–59	Second Class
40–49	Third Class

Condition 5 - Scholarship

Display:
Scholarship: Eligible
only when:
percentage >= 85 AND every subject >= 75
Otherwise:
Scholarship: Not Eligible

Sample Output

```
===== STUDENT RESULT SYSTEM =====
Enter student name: Rahul
Enter Java marks: 85
Enter Python marks: 78
Enter Database marks: 92
Enter Web Technology marks: 88
Enter Aptitude marks: 81
===== RESULT =====
Student Name : Rahul
Total        : 424 / 500
Percentage   : 84.8%
Result       : PASS
Grade        : A
Division     : First Class
Scholarship  : Not Eligible
```

Bank Management System

Create a **console-based Bank Management System** in Python. The program should allow a user to deposit money, withdraw money, check the current balance, and exit the application.

Initial Balance

The account should start with an initial balance of **₹5,000**.

Main Menu

When the program starts, display the following menu:

===== BANK MANAGEMENT SYSTEM =====

1. Deposit
2. Withdraw
3. View Balance
4. Exit

The menu should continue to appear until the user selects **Exit**.

Use if and loop if necessary

Expected Output

```
===== BANK MANAGEMENT SYSTEM =====
1. Deposit
2. Withdraw
3. View Balance
4. Exit

Enter your choice: 1
Enter deposit amount: 2000
Amount deposited successfully.
New Balance: 7000.0

===== BANK MANAGEMENT SYSTEM =====
1. Deposit
2. Withdraw
3. View Balance
4. Exit

Enter your choice: 4
Thank you for using the Bank Management System.
```